

CDACL-002

Customer Segmentation Analysis

K-Means Clustering on Shopping Trends Data — MySQL · Orange · Power BI

Prepared by
Supratik Chattopadhyay

Dataset: Shopping Trends (N = 3,900 customers)

1. Executive Summary

This report presents the results of a K-Means clustering analysis performed on a customer shopping-trends dataset (N = 3,900) to identify distinct customer segments for targeted business strategy. Using Age, Purchase Amount, Previous Purchases, and Review Rating as clustering features, three statistically distinct customer segments were identified — each validated using one-way ANOVA ($p < 0.001$ across all four variables).

The analysis was carried out using Orange Data Mining for clustering and visualization, with MySQL for data preparation and Power BI reserved for the final dashboard presentation.

- C1 — New / Low-Engagement customers: younger, moderate spend, very low purchase history.
- C2 — Frequent, Price-Sensitive customers: high purchase frequency but the lowest average spend and satisfaction.
- C3 — High-Value / VIP customers: highest spend, high frequency, and the highest satisfaction ratings.

2. Methodology

Customer records were loaded and pre-processed, then passed through a K-Means clustering pipeline ($k = 3$) built in Orange Data Mining. The resulting cluster assignments were evaluated in parallel using a Data Table, Box Plot, and Scatter Plot widget to validate separation quality.

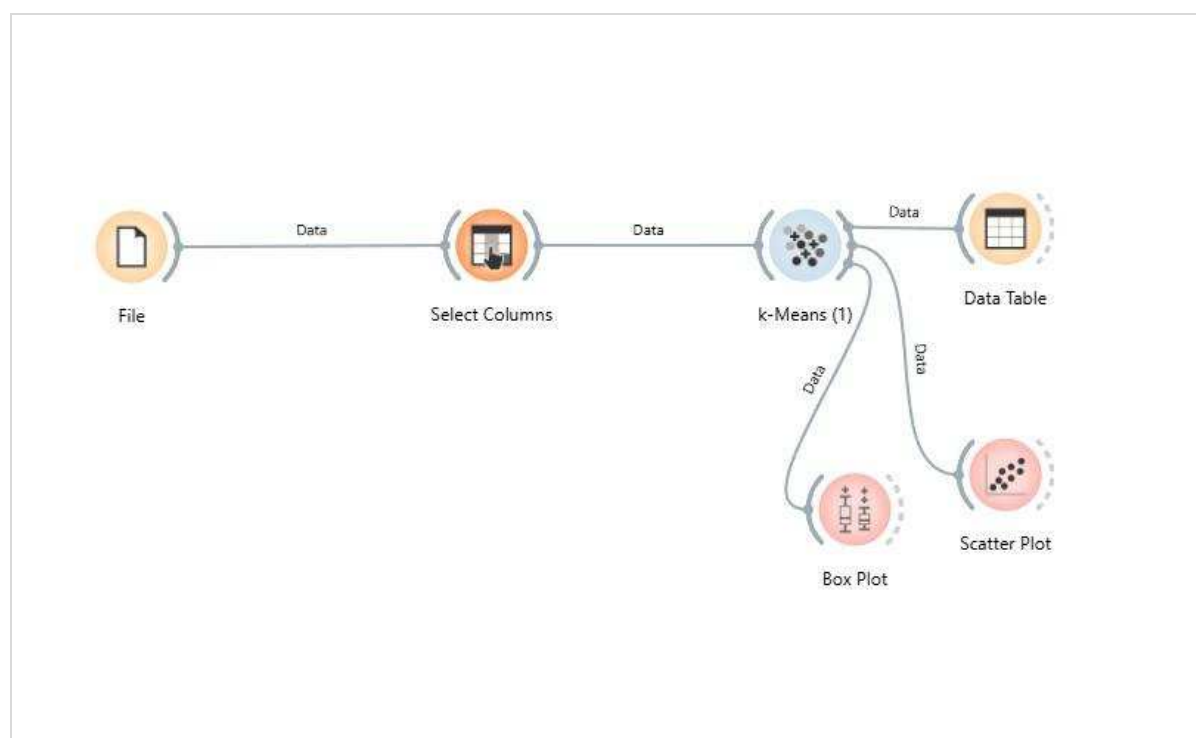


Figure 1 — Orange Data Mining workflow: File → Select Columns → k-Means → {Data Table, Box Plot, Scatter Plot}

Each numeric feature (Age, Purchase Amount, Previous Purchases, Review Rating) was compared across the three resulting clusters using one-way ANOVA to confirm the segments are statistically distinct rather than arbitrary groupings.

3. Cluster Comparison — Statistical Validation

The following box plots compare each clustering variable across the three segments. All four ANOVA tests returned $p = 0.000$, confirming the clusters differ significantly on every dimension used.

3.1 Age Distribution by Cluster

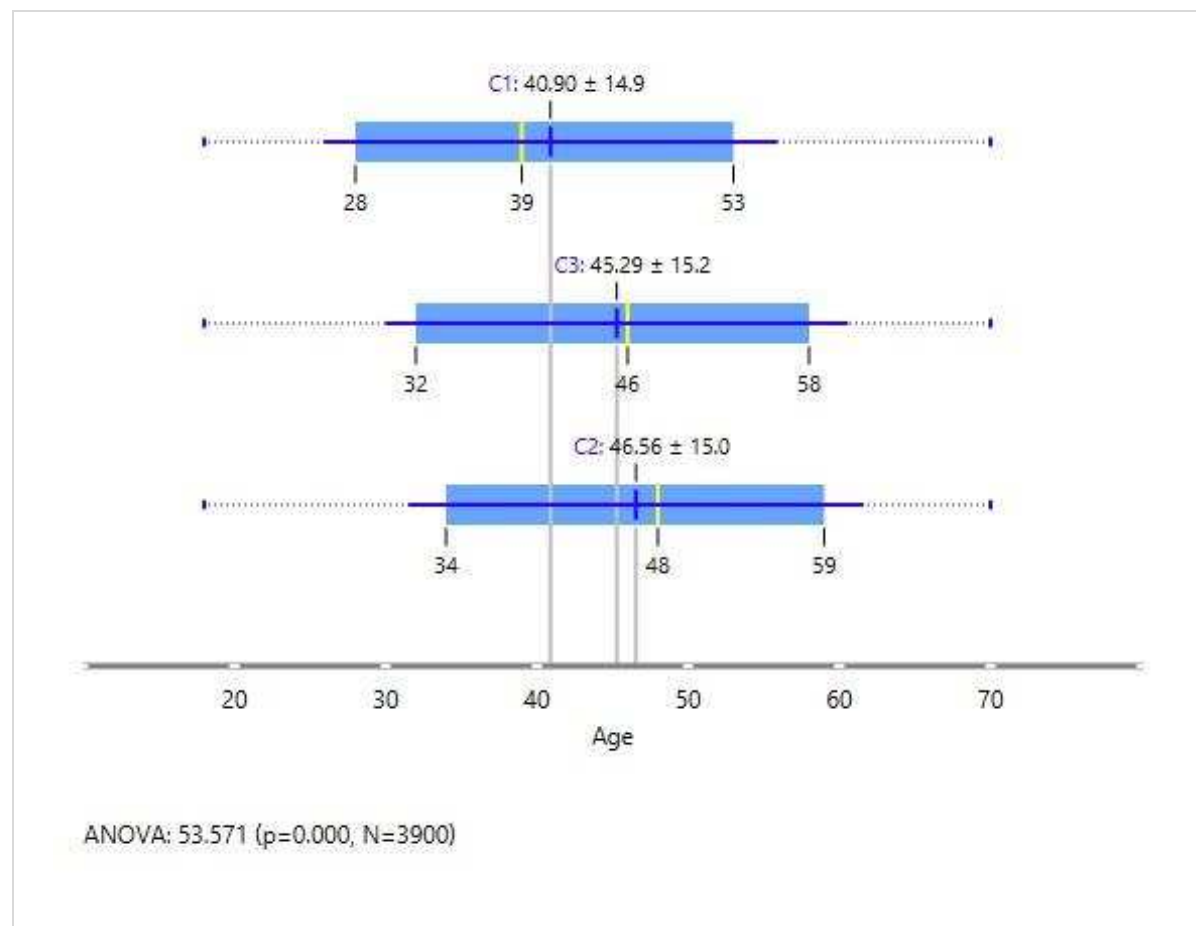


Figure 2 — Age by Cluster | ANOVA $F = 53.571$, $p = 0.000$, $N = 3,900$

3.2 Purchase Amount (USD) by Cluster

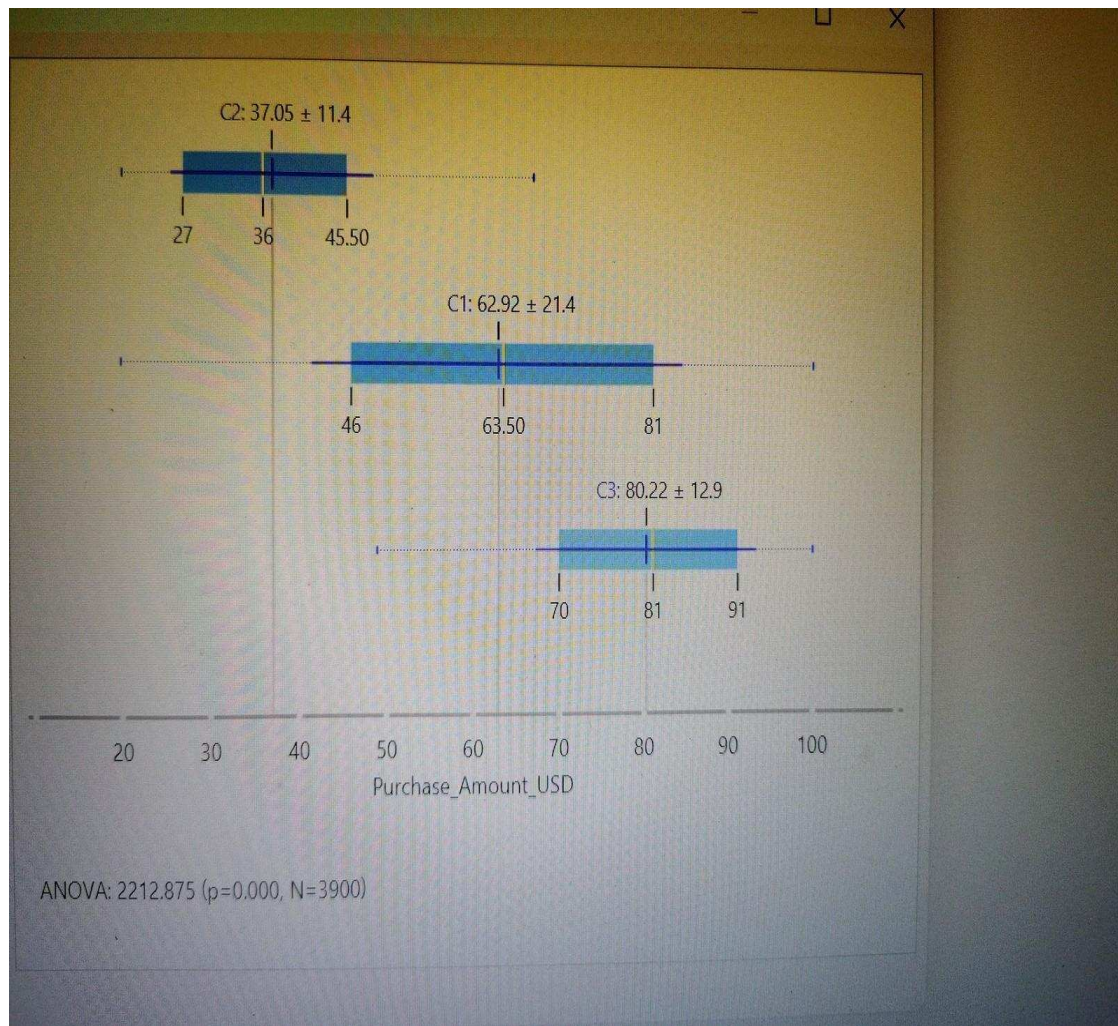


Figure 3 — Purchase Amount (USD) by Cluster | ANOVA $F = 2212.875$, $p = 0.000$, $N = 3,900$

3.3 Previous Purchases by Cluster

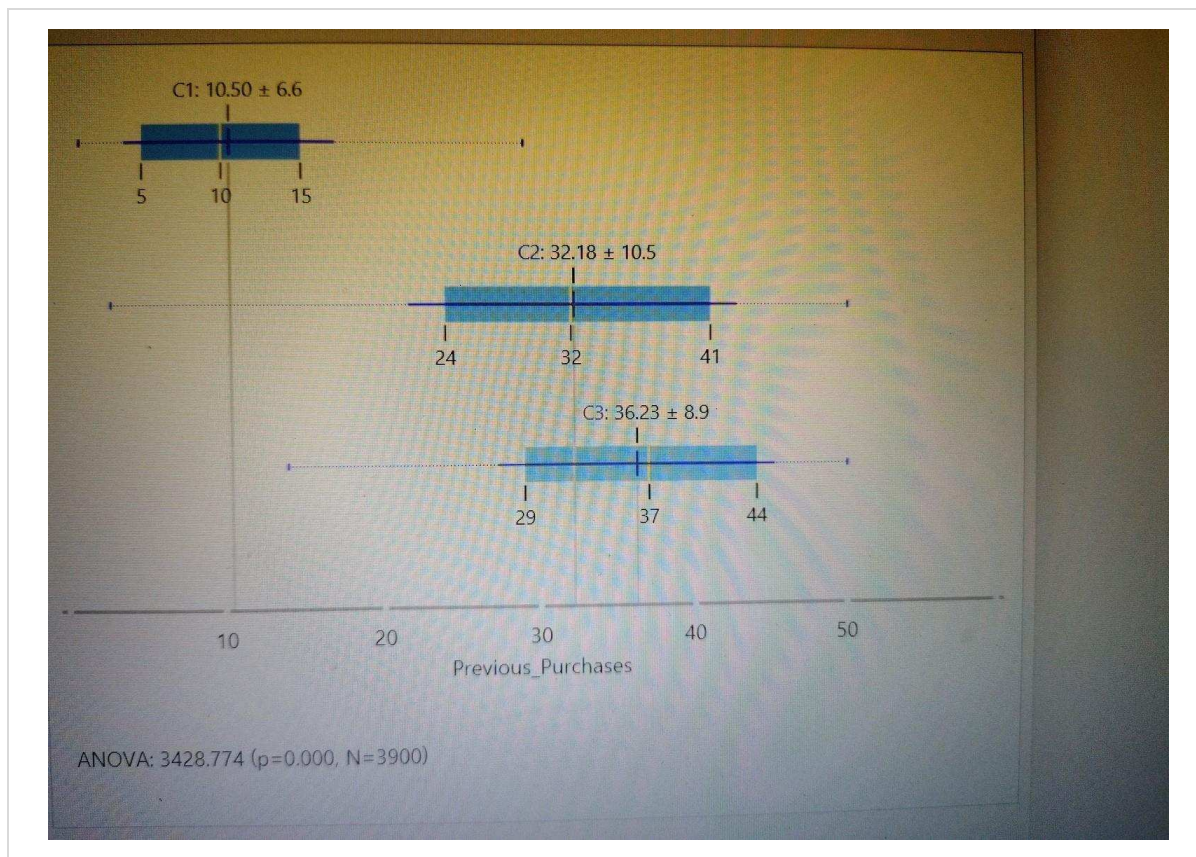


Figure 4 — Previous Purchases by Cluster | ANOVA $F = 3428.774$, $p = 0.000$, $N = 3,900$

3.4 Review Rating by Cluster

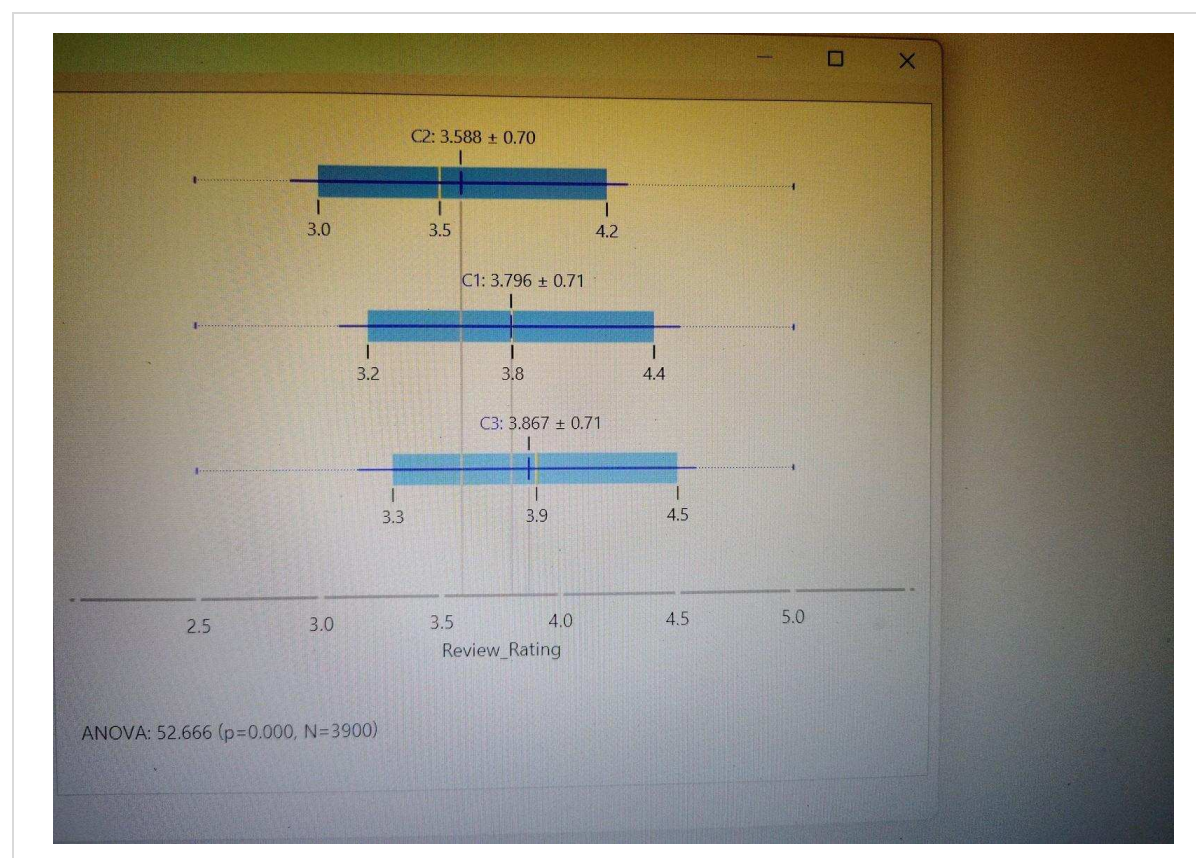


Figure 5 — Review Rating by Cluster | ANOVA $F = 52.666$, $p = 0.000$, $N = 3,900$

4. Cluster Profile Summary

The table below consolidates the mean $\pm$ standard deviation for each clustering variable, by segment.

Metric	C1 (New / Low-Engagement)	C2 (Frequent, Price-Sensitive)	C3 (High-Value / VIP)
Age (yrs)	40.90 $\pm$ 14.9	46.56 $\pm$ 15.0	45.29 $\pm$ 15.2
Purchase Amount (USD)	62.92 $\pm$ 21.4	37.05 $\pm$ 11.4	80.22 $\pm$ 12.9
Previous Purchases	10.50 $\pm$ 6.6	32.18 $\pm$ 10.5	36.23 $\pm$ 8.9
Review Rating (/5)	3.796 $\pm$ 0.71	3.588 $\pm$ 0.70	3.867 $\pm$ 0.71

ANOVA significance summary for all four clustering variables:

Variable	ANOVA F-statistic	p-value	Significant?
Age	53.571	0.000	Yes

Variable	ANOVA F-statistic	p-value	Significant?
Purchase Amount (USD)	2212.875	0.000	Yes
Previous Purchases	3428.774	0.000	Yes
Review Rating	52.666	0.000	Yes

5. Cluster Separation — Visual Validation

Plotting Purchase Amount against Previous Purchases shows the three clusters occupying distinct spatial regions, confirming that the K-Means solution produces well-separated, interpretable segments rather than overlapping noise.

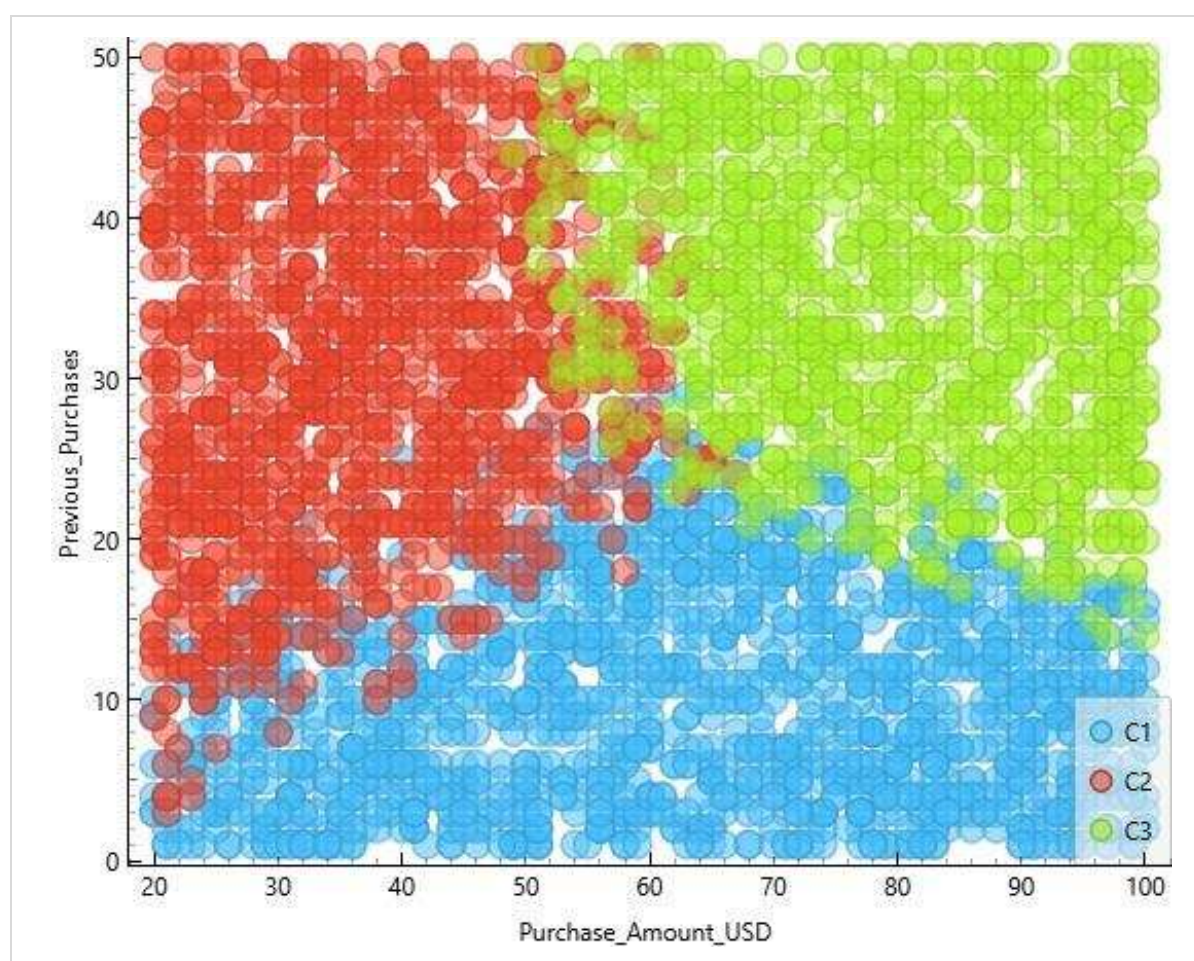


Figure 6 — Purchase Amount vs. Previous Purchases, coloured by cluster (C1 = blue, C2 = red, C3 = green)

- C1 (blue) occupies the low-spend / low-frequency region.
- C2 (red) occupies the low-to-mid spend / high-frequency region.
- C3 (green) occupies the high-spend region across all frequency levels.
- A small overlap between C2 and C3 appears around \$50–\$60 spend, consistent with their close review-rating averages (3.588 vs 3.867).

6. Cluster Interpretation & Business Recommendations

C1 — New / Low-Engagement Customers

Youngest average age (40.9 yrs), moderate spend (~\$63), but by far the lowest purchase history (~10.5 prior purchases). Likely newer or infrequent shoppers still building a relationship with the brand.

- Recommendation: onboarding incentives, welcome discounts, and engagement campaigns to increase purchase frequency.

C2 — Frequent, Price-Sensitive Customers

Highest purchase frequency alongside the lowest average spend (~\$37) and lowest review rating (3.588). These are loyal-by-habit shoppers who may be price-driven rather than brand-driven.

- Recommendation: loyalty/bundle pricing to lift basket size, and investigate the satisfaction gap driving lower ratings.

C3 — High-Value / VIP Customers

Highest spend (~\$80), high frequency (~36 purchases), and the highest satisfaction (3.867). This is the most valuable and most loyal segment.

- Recommendation: VIP retention programs, early access, and premium loyalty tiers to protect this high-margin segment.

7. Notes & Next Steps

- A Silhouette Score / Elbow analysis should be documented to formally justify $k = 3$.
- Confirm feature normalization was applied before K-Means, since Purchase Amount and Review Rating are on very different numeric scales.
- A post-hoc Tukey HSD test is recommended to confirm all pairwise cluster differences are significant, particularly for the closer C2/C3 review-rating means.
- Final segment definitions to be carried into the Power BI dashboard for stakeholder presentation.